

Class Activity- Exception Handling

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

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Class Activity- Exception Handling

QUESTIONS

Q1
ArrayIndexOutOfBoundsException
10 marks

Q2
calculator application
10 marks

Q3
student registration system
10 marks

3/3 answered

Q1 ArrayIndexOutOfBoundsException 10 marks

You are developing a student grade management system. Write a program that stores 5 student grades in an array and allows users to retrieve a grade by entering the student's position (1-5). Handle cases where the position is out of range or input is invalid

Sample Output : 15

Your Code

```
1 import java.util.Scanner;
2 import java.util.NoSuchElementException;
3 public class StudentGradeManagement {
4     public static void main(String[] args) {
5         int[] grades = {85, 90, 78, 92, 15};
6         Scanner scanner = new Scanner(System.in);
7         try {
8             System.out.print("Enter student position (1-5): ");
9             int position = Integer.parseInt(scanner.nextLine().trim());
10            int index = position - 1;
11            int grade = grades[index];
12            System.out.println("Grade for student at position " + position + " is: " + grade);
13        } catch (ArrayIndexOutOfBoundsException e) {
14            System.out.println("Error: Position out of range. Please enter a valid position (1-5).");
15        } catch (NumberFormatException e) {
16            System.out.println("Error: Invalid input. Please enter a valid integer (1-5).");
17        }
18    }
19 }
```

Input

5

Output

Enter student position (1-5): Grade for student at position 5 is: 15

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3/3 answered

Q2 calculator application 10 marks

You are building a simple calculator application. Write a program that takes two integers as input and divides them. Handle the case where the denominator is zero.

Your Code

```
1 import java.util.Scanner;
2 import java.util.InputMismatchException;
3 import java.util.NoSuchElementException;
4 public class Calculator {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         try {
9             System.out.print("Enter numerator: ");
10            int numerator = sc.nextInt();
11
12            System.out.print("Enter denominator: ");
13            int denominator = sc.nextInt();
14
15            int result = numerator / denominator;
16            System.out.println("Result: " + result);
17        } catch (ArithmeticException e) {
18            System.out.println("Error: Division by zero is not allowed.");
19        }
20    }
21 }
```

Input

3 3

Output

Enter numerator: Enter denominator: Result: 1

Class Activity- Exception Handling

← Back

QUESTIONS

Q1 ✓
ArrayIndexOutOfBoundsException
10 marks**Q2** ✓
calculator application
10 marks**Q3** ✓
student registration system
10 marks

3/3 answered

Q3 student registration system

10 marks

You are developing a student registration system for a school. Students must be between 5 and 18 years old to enroll. Create a custom exception called `InvalidAgeException` that is thrown when a student's age is outside the valid range. The program should allow the user to register multiple students and display their information.

Your Code

```
1 import java.util.Scanner;
2
3 public class InvalidAgeException extends Exception {
4
5     InvalidAgeException(String message) {
6         super(message);
7     }
8
9     public static void main(String[] args) {
10
11         Scanner sc = new Scanner(System.in);
12
13         int n = sc.nextInt();
14
15         for (int i = 0; i < n; i++) {
```

Input

```
3
John
15
Timmy
```

Output

```
Student registered successfully
Name: John
Age: 15
Invalid age for student: Timmy
Student registered successfully
Name: Timmy
```